

SYDNEY BOYS HIGH SCHOOL



HSC TRIAL EXAMINATION 2006

PHYSICS

General Instructions

Reading Time 5 minutes
 Working Time 3 hours
 Write using blue or black pen
 Draw diagrams using pencil.
 Board- approved calculators may be used.
 A data sheet, formulae sheets and
 Periodic Table are provided with this
 paper.

Marks may be allocated to working
 Show all working

Total marks – 100**Section I** Pages 2 – 16**Total marks 75**

This section has two parts, Part A and
 Part B.

Part A – 15 marks

- Questions 1 – 15
- Allow about 30 minutes for this
part

Part B – 60 marks

- Attempt Questions 16 – 30
- Allow about hour and 45 minutes
for this part.

Section II Page 15**Total marks 25**

1. There are a number of reasons which contribute to variations in the value of the acceleration due to gravity at specific locations on the surface of the Earth.

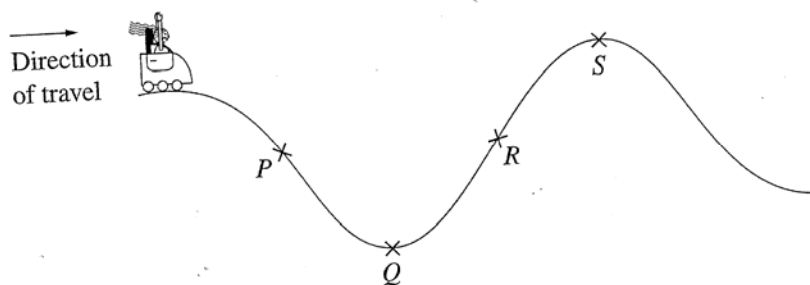
Which of the following pairs of reasons would **not** be responsible for such variations?

- (A) Crustal variations and the shape of the Earth.
- (B) The shape of the Earth and the height above sea level.
- (C) Height above sea level and the Earth's spin.
- (D) Crustal variations and the Earth's orbit around the Sun.

2. The Earth, of radius and mass, 6.38×10^6 m and 5.98×10^{24} kg respectively, has an artificial satellite. The satellite orbits at an altitude of 300 km, has a mass of 200 kg and travels with a velocity of $20\,000 \text{ km h}^{-1}$. The gravitational force acting on the satellite is;

- (A) $1.8 \times 10^{-3} \text{ N}$
- (B) $2.3 \times 10^{-1} \text{ N}$
- (C) $1.8 \times 10^3 \text{ N}$
- (D) $2.3 \times 10^3 \text{ N}$

3. The diagram shows four positions of a car on a roller coaster ride.



At which point during this ride would the occupant experience maximum 'g force'?

- (A) P
- (B) Q
- (C) R
- (D) S

4. The table contains information related to two planets orbiting a distant star.

Planets	Mass (kg)	Orbital radius (m)	Radius of planets (m)	Length of day (s)	Orbital period (s)
Alif	1.21×10^{25}	4.00×10^{11}	8.0×10^6	9.5×10^4	8.75×10^7
Ba	1.50×10^{24}	8.00×10^{11}	4.0×10^6	4.7×10^4	—

The orbital period of the planet Ba can be determined by using data selected from this table.

What is the orbital period of the planet Ba?

- (A) $3.10 \times 10^7 \text{ s}$
- (B) $5.51 \times 10^7 \text{ s}$
- (C) $1.39 \times 10^8 \text{ s}$
- (D) $2.47 \times 10^8 \text{ s}$

5. A radioactive particle used in a linear accelerator. Measured at rest relative to the laboratory it has a half life of $2.5\mu\text{s}$. When measured at constant speed by an observer in the laboratory, its half life has increased to $10\mu\text{s}$.

What is the speed of the particle relative to the laboratory?

- (A) $1.68 \times 10^8 \text{ ms}^{-1}$
- (B) $2.10 \times 10^8 \text{ ms}^{-1}$
- (C) $290\,000\,000 \text{ ms}^{-1}$
- (D) $2.60 \times 10^8 \text{ ms}^{-1}$

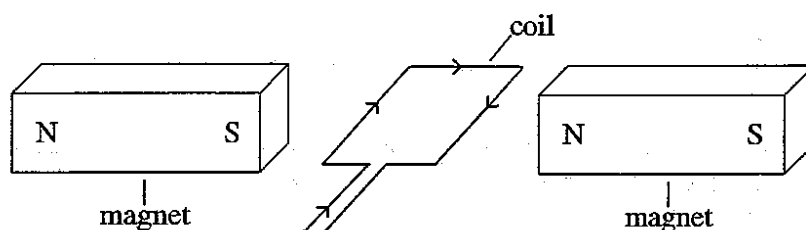
6. Identify which of the following is **not** a component of a DC generator.

- (A) brushes
- (B) coil
- (C) magnetic field
- (D) slip-rings

7. Two long parallel wires are carrying electrical currents. The direction of the current in one of the wires is reversed. How does this affect the force between the wires?

- (A) The force does not change.
- (B) The force changes direction.
- (C) The force increases.
- (D) The force decreases.

8. The following diagram shows a coil of wire between two magnets.



When a current passes through the coil in the direction shown, which is now free to move, the coil will:

- (A) start rotating clockwise (viewed from the front)
- (B) not move
- (C) move vertically
- (D) start rotating anticlockwise (viewed from front)

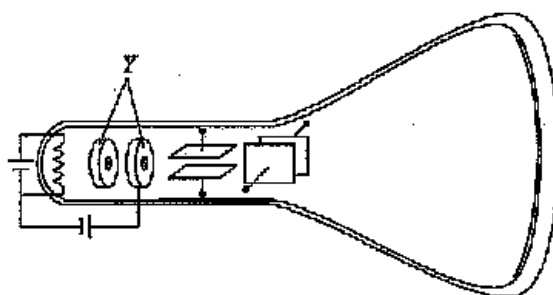
9. A piece of wire 1.0cm long is at right angles to a magnetic field whose magnetic flux density is 1.5T . A current of 2.0 A flows in the wire. What is the magnitude of the force on the wire?

- (A) 3.0 N
- (B) 0.3 N
- (C) 0.03 N
- (D) zero

10. A transformer is needed to convert an input voltage of 6000V to an output voltage of 240V. The **type of transformer** and the **ratio** of the number of turns in its secondary coil to the number of turns in its primary coil are

- (A) step up, 25:1
- (B) step up, 1:25
- (C) step down, 25:1
- (D) step down, 1:25

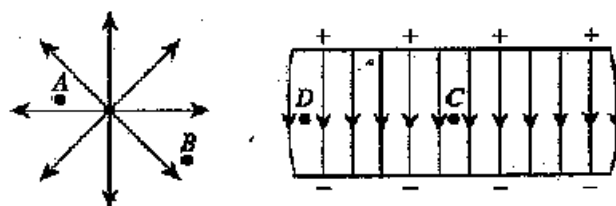
11. The following diagram shows a simple cathode ray tube from an oscilloscope.



The parts labelled Y have the function of

- (A) producing electrons.
- (B) showing the position of the beam.
- (C) deflecting the beam horizontally.
- (D) accelerating the electrons.

12. The following diagram shows the electric fields near a point charge and between parallel plates.



At which point is the magnitude of the electric field greatest?

- (A) A
- (B) B
- (C) C
- (D) D

13. Which of the following statements is correct?

- (A) Einstein was the first person to observe the photoelectric effect.
- (B) Planck hypothesised that energy was exchanged, in quanta amounts, by the atomic oscillators of a black body.
- (C) Hertz performed experiments to measure the speed of light, using radio waves.
- (D) Einstein predicted that for a black body, as the wavelength shortens, the radiation intensity will increase.

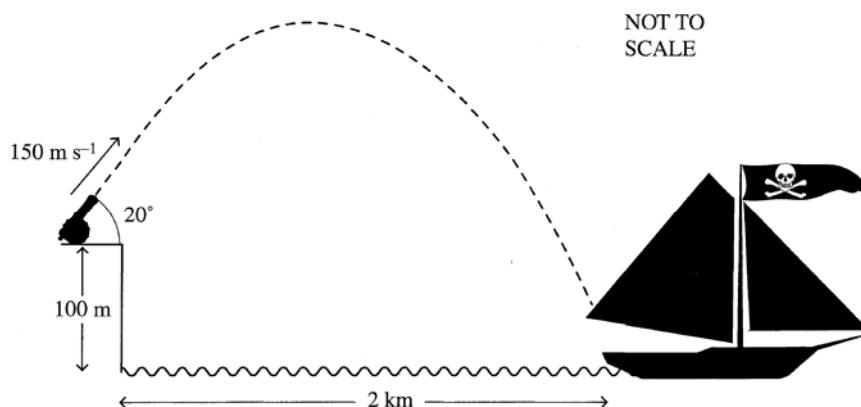
14. Solid state devices replaced thermionic devices because thermionic devices;

- (A) were much smaller and required less current in their circuits.
- (B) had a much longer life as they did not become warm at all.
- (C) were less reliable and were easily broken.
- (D) allowed the cathode coating to evaporate, helping current flow.

15. In metal conductors, the resistance

- (A) increases as the amount of impurities increase.
- (B) increases as the lattice vibrations decrease.
- (C) decreases as more electrons are scattered by lattice vibrations.
- (D) decreases as the temperature of the metal increases.

16.



An enemy ship was sailing 2km from the coast. A cannon on a 100 metre-high cliff fired a projectile at an angle of 20° to the horizontal, at a speed of 150 m/s.

(a) Determine the vertical and horizontal components of the initial velocity.

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(b) Calculate the time taken for the cannon ball to reach the maximum height **and** the maximum height of the cannon ball above the water.

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(c) Calculate

(i) the range of the cannon ball

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(ii) how far from the ship the cannon ball landed **1**

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(d) Describe an adjustment of the cannon that is necessary for a cannon ball to hit the ship. **1**

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17. Explain why all low earth orbit satellites will eventually fall to the Earth's surface. **2**

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18. A boy standing on train station platform observes a NLST (near light-speed transport) train pass through the station. He observes the clocks on the train to be running slower than normal. However, a girl on the train observes the boy's watch, and notices that his watch is running slower than the clocks on the train.

Account for the above situation with reference to the principle of relativity. **3**

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19. Michelson and Morley set up an experiment to measure the velocity of Earth relative to the aether. **Mark**

(a) Outline TWO features of the aether model for the transmission of light. 2

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(b) Recount the Michelson and Morley experiment, which attempted to measure the relative velocity of Earth through the aether, and describe the results they anticipated. 4

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20. In your course you performed an investigation to demonstrate the production of an alternating current. **Mark**

(a) Describe an experiment you did to produce alternating current, with particular reference to how you verified that alternating current was actually produced.

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(b) Describe two advantages of using AC generators for large-scale electrical power production. 2

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21. The photograph shows a small electrical motor from an electric drill.

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(a) Name the labelled parts
A, B, and C **and**

(b) Describe the function
Of each

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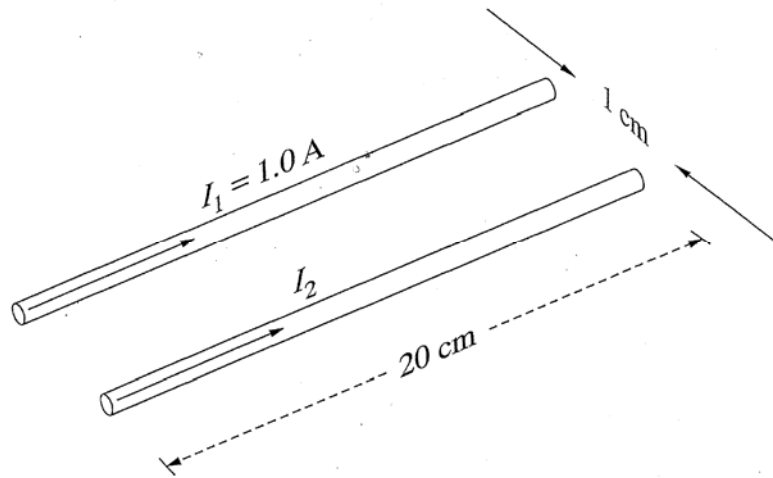
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22. The diagram shows part of an experiment designed to measure the force between two parallel current-carrying conductors.



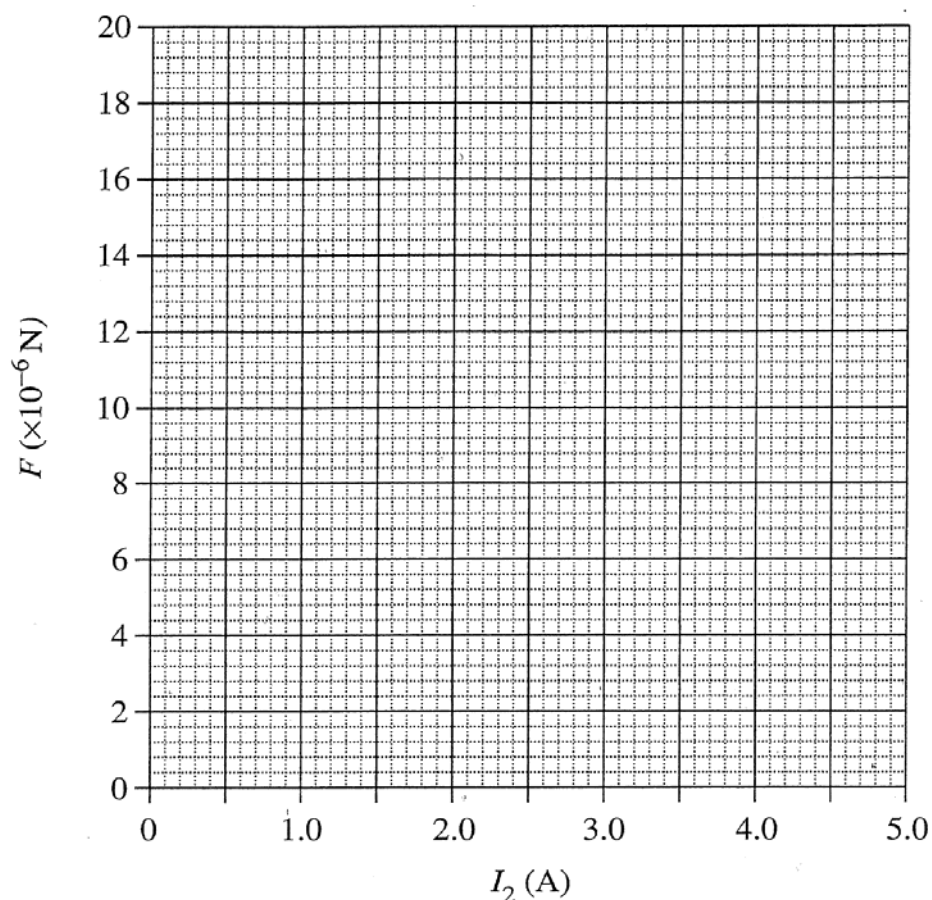
The experimental results are tabulated below.

$I_2 \text{ (A)}$	Force ($\times 10^{-6} \text{ N}$)
0	0
2.0	7
3.0	11
4.0	14
5.0	18

Question 22 (continued)

Mark
3

(a) Plot the data and draw the line of best fit.



(b) Calculate the gradient of the line of best fit for the graph.

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(c) Write an expression for the magnetic force constant k in the terms of the gradient and other variables.

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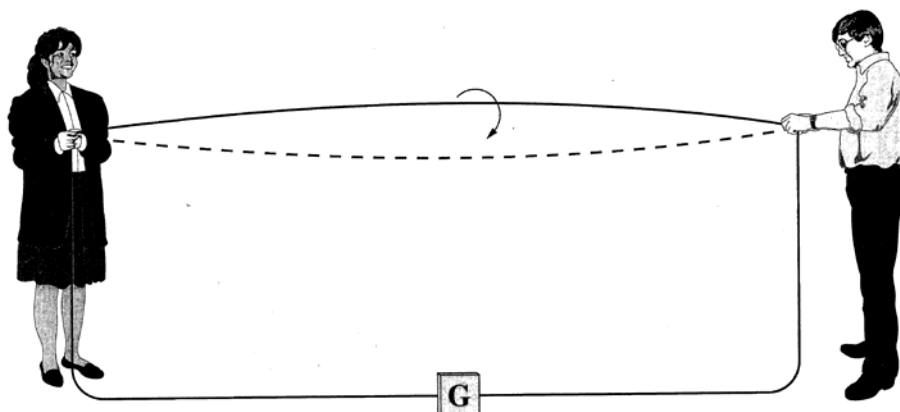
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(d) Use this expression and the gradient calculated in part (b) to determine the value of the magnetic force constant k .

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23. In a particular experiment a long length of copper wire of very resistance is rotated by two students. The ends of the wire are connected to a galvanometer G, and a current is detected.



Explain the effect of increasing the speed of rotation on the current measured by the galvanometer **4**

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24. Using silicon as an example of a semiconductor, describe how it carries a current and how doping effects the process. **2**

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25. During your course you carried out an investigation to model behaviour of semiconductors, including the concept of holes. **Marks 3**

Outline what you did in your investigation. Explain how the model showed conduction in semiconductors.

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26. In early studies, the observed characteristics of cathode rays led to the belief that they were electromagnetic waves. **3**

Describe the wave-like properties of cathode rays and explain how other evidence shows them to be particles.

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27. A physics student was conducting an investigation on the photoelectric effect. The student used an infrared laser with a wavelength of $1.55 \times 10^{-6}\text{m}$ for this investigation.

(a) Calculate the energy of a photon from this laser. **2**

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(b) When the laser light was shone onto a photo-cell, no current was detected. The student increased the intensity of the light but still detected no current. **3**
Explain this observation.

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28. (a) Calculate the frequency of a photon of blue light of wavelength 460nm. **1**

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(b) Identify Planck's hypothesis that allowed him to successfully account for the black body radiation curve. **1**

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29. Outline how Hertz measured the speed of radio waves. **3**

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30. Describe how superconductors and magnetic levitation have played a part in the development of the maglev train. **2**

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SECTION II- Option

From Quanta to Quarks. (25 Marks)

Marks

(a) Discuss Rutherford's model of the nuclear atom with orbiting electrons 4

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(b) In refining the model of the atom, Bohr began with three postulates. 2
State 2 of Bohr's postulates.

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(c) Identify experimental evidence that supported one of Bohr's postulates and explain 2
how it provided this support.

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(d) Define the term transmutation. 1

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Marks

(e) As a result of the studying the electrons emitted during beta decay, Pauli suggested the 4
existence of a then unknown particle. **Discuss** Pauli's suggestion, and relate this to the energy

of the emitted electrons.

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(f) Calculate the mass of a particle that has a De Broglie wavelength of 2.5×10^{-12} m when moving with speed of 200ms^{-1} . 1

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(g) Calculate the energy of the lowest frequency photon emitted in the Balmer series? 2

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(h) Write the transmutation equation for the beta decay of Bi^{210}_{83} . 2

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Marks

(i) Explain the significance of the conservation laws, in Chadwick's discovery of the neutron. 4

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(j) Justify the existence of the strong nuclear force.

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END OF EXAMINATION

Data sheet

Charge on the electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Magnetic force constant, $\left(k \equiv \frac{\mu_0}{2\pi}\right)$	$2 \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth	$6.0 \times 10^{24} \text{ kg}$
Planck constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg constant, R_{hydrogen}	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Formulae sheet

$$v = f\lambda$$

$$I \propto \frac{1}{d^2}$$

$$\frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

$$E_p = -G \frac{m_1 m_2}{r}$$

$$F = mg$$

$$v_x^2 = u_x^2$$

$$v = u + at$$

$$E = \frac{F}{q}$$

$$R = \frac{V}{I}$$

$$P = VI$$

$$\text{Energy} = VIt$$

$$v_y^2 = u_y^2 + 2a_y \Delta y$$

$$\Delta x = u_x t$$

$$\Delta y = u_y t + \frac{1}{2} a_y t^2$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$v_{av} = \frac{\Delta r}{\Delta t}$$

$$a_{av} = \frac{\Delta v}{\Delta t} \text{ therefore } a_{av} = \frac{v - u}{t}$$

$$\Sigma F = ma$$

$$F = \frac{mv^2}{r}$$

$$E_k = \frac{1}{2} mv^2$$

$$W = Fs$$

$$p = mv$$

$$\text{Impulse} = Ft$$

$$F = \frac{Gm_1 m_2}{d^2}$$

$$E = mc^2$$

$$l_v = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$t_v = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$m_v = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Formulae sheet

$$\frac{F}{l} = k \frac{I_1 I_2}{d}$$

$$d = \frac{1}{p}$$

$$F = BIl \sin \theta$$

$$M = m - 5 \log \left(\frac{d}{10} \right)$$

$$\tau = Fd$$

$$\frac{I_A}{I_B} = 100^{(m_B - m_A)/5}$$

$$\tau = nBIA \cos \theta$$

$$m_1 + m_2 = \frac{4\pi^2 r^3}{GT^2}$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

$$F = qvB \sin \theta$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$E = \frac{V}{d}$$

$$\lambda = \frac{h}{mv}$$

$$E = hf$$

$$c = f\lambda$$

$$A_0 = \frac{V_{\text{out}}}{V_{\text{in}}}$$

$$Z = \rho v$$

$$\frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{R_f}{R_i}$$

$$\frac{I_r}{I_o} = \frac{[Z_2 - Z_1]^2}{[Z_2 + Z_1]^2}$$

Periodic Table of the Elements

1		2	
1	H 1.008 Hydrogen	2	He 4.003 Helium
KEY Atomic number Symbol of element Name of element			
3	Li 6.941 Lithium	4	Be 9.012 Beryllium
11	Na 22.99 Sodium	12	Mg 24.31 Magnesium
19	K 39.10 Potassium	20	Ca 40.08 Calcium
37	Rb 85.47 Rubidium	38	Sr 87.62 Strontium
55	Cs 132.9 Caesium	56	Ba 137.3 Barium
87	Fr [223.0] Francium	88	Ra [226.0] Radium
21	Sc 44.96 Scandium	22	Ti 47.87 Titanium
39	Y 88.91 Yttrium	40	Zr 91.22 Zirconium
57-71	Lanthanides	72	Hf 178.5 Hafnium
89-103	Actinides	104	Rf [261.1] Rutherfordium
23	V 50.94 Vanadium	24	Cr 52.00 Chromium
41	Nb 92.91 Niobium	42	Mo 95.94 Molybdenum
73	Ta 180.9 Tantalum	74	W 183.8 Tungsten
105	Db [262.1] Dubnium	106	Sg [266.1] Seaborgium
107	Bh [264.1] Bohrium	108	Hs [277] Hassium
109	Mt [268] Meitnerium	110	Ds [271] Darmstadtium
111	Rg [272] Roentgenium		
25	Mn 54.94 Manganese	26	Fe 55.85 Iron
43	Tc [98.91] Technetium	44	Ru 101.1 Ruthenium
75	Re 186.2 Rhenium	76	Os 190.2 Osmium
101	La [137.3] Lanthanum	102	Hf 178.5 Hafnium
103	Pr 140.9 Praseodymium	104	Zr 91.22 Zirconium
105	Nd 144.9 Neodymium	106	Mo 95.94 Molybdenum
107	Pm [146.9] Promethium	108	Tc [98.91] Technetium
109	Sm 150.4 Samarium	110	Ru 101.1 Ruthenium
111	Eu 152.0 Europium	112	Rh 102.9 Rhodium
113	Gd 157.3 Gadolinium	114	Pd 106.4 Palladium
115	Tb 158.9 Terbium	116	Ag 107.9 Silver
117	Dy 162.5 Dysprosium	118	Cd 112.4 Cadmium
119	Ho 164.9 Holmium	120	In 114.8 Indium
121	Er 167.3 Erbium	122	Sn 118.7 Tin
123	Tm 168.9 Thulium	124	Pb 207.2 Lead
125	Yb 173.0 Ytterbium	126	Bi 209.0 Bismuth
127	Lu 175.0 Lutetium	128	Po [209.0] Polonium
129		130	At [210.0] Astatine
131		132	Rn [222.0] Radon

57	La 138.9 Lanthanum	58	Ce 140.1 Cerium	59	Pr 140.9 Praseodymium	60	Nd 144.9 Neodymium	61	Pm [146.9] Promethium	62	Sm 150.4 Samarium	63	Eu 152.0 Europium	64	Gd 157.3 Gadolinium	65	Tb 158.9 Terbium	66	Dy 162.5 Dysprosium	67	Ho 164.9 Holmium	68	Er 167.3 Erbium	69	Tm 168.9 Thulium	70	Yb 173.0 Ytterbium	71	Lu 175.0 Lutetium
89	Ac [227.0] Actinium	90	Th 232.0 Thorium	91	Pa 231.0 Protactinium	92	U 238.0 Uranium	93	Np [237.0] Neptunium	94	Pu [244.1] Plutonium	95	Am [243.1] Americium	96	Cm [247.1] Curium	97	Bk [247.1] Berkelium	98	Cf [251.1] Californium	99	Es [252.1] Einsteinium	100	Fm [257.1] Fermium	101	Md [258.1] Mendelevium	102	No [259.1] Nobelium	103	Lr [262.1] Lawrencium

Where the atomic weight is not known, the relative atomic mass of the most common radioactive isotope is shown in brackets.

The atomic weights of Np and Tc are given for the isotopes ^{237}Np and ^{99}Tc .

SYDNEY BOYS HIGH SCHOOL **HSC PHYSICS TRIAL** **MULTIPLE CHOICE ANSWER SHEET**

Student number

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12				
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14				
15				

ANSWERS

Student Number.....

FINAL VERSION

SYDNEY BOYS HIGH SCHOOL
HSC PHYSICS TRIAL
MULTIPLE CHOICE ANSWER SHEET



Student number

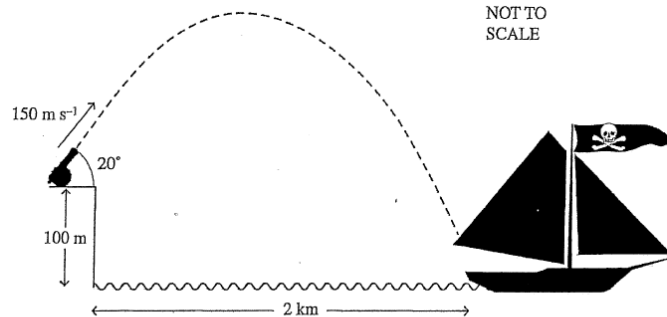
	A	B	C	D
1				X
2			X	
3		X		
4	WAV			X
5			X	WAV
6				X
7		X		
8	X			
9			X	
10				X
11				X
12			X	
13		X		
14			X	
15	X			

Changes
A → D
D → C

FINAL VERSION

Section B (60 marks) Show all working
16.

Marks



An enemy ship was sailing 2 km from the coast. A cannon on a 100 metre-high cliff fired a canon ball at an angle of 20° to the horizontal, at speed of 150 m/s.

- (a) Determine the vertical and horizontal components of the initial velocity.

$$\mu_x = 150 \cos 20 = 141 \text{ m/s} \quad 2$$

$$\mu_y = 150 \sin 20 = 51.3 \text{ m/s}$$

- (b) Calculate the time taken for the cannon ball to reach the maximum height and hence the maximum height of the cannon ball above the water. 3

$$v_y = 0, a = -9.8, \mu_y = 51.3 \text{ m/s}$$

$$v = u + at$$

$$t = \frac{v_y - \mu_y}{a}$$

$$= \frac{0 - 51.3}{-9.8}$$

$$= 5.24 \text{ s}$$

$$\Delta y = \mu_y t + \frac{1}{2} a t^2$$

$$= 51.3 \times 5.24 + \frac{1}{2} \times -9.8 \times 5.24^2$$

$$= 134.27 \text{ m}$$

$$\therefore \text{Height above water} \\ = 134.27 + 100 \\ = 234 \text{ m}$$

- (c) Calculate the range of the cannon ball and hence determine how far from the ship the cannon ball landed. Marks

12/3

More $h = 234.27 \text{ m}$

time to fall from 234.27 m

$$t = \sqrt{\frac{2 \times \Delta y}{a_y}}$$

$$= \sqrt{\frac{2 \times 234.27}{9.8}}$$

$$= 6.91 \text{ s}$$

$$\therefore \text{TOTAL TIME} = 6.91 + 5.24 = 12.2 \text{ s}$$

$$\text{Range } \Delta x = u \times t$$

$$= 140.95 \times 12.15$$

$$= 1713.2$$

$$= 1713 \text{ m}$$

2 marks

Cannon ball landed 287

m short of the ship 1 mark

- (d) Describe an adjustment of the cannon that is necessary for a cannon ball to the ship. 1

- Increase angle (to horizontal) of launch ("adjust" not good enough)
(or increase gun powder to give more velocity)

17. Explain why all low-Earth satellites will eventually fall to the Earth's surface. 2

LEO Satellites (300-1500 km altitude) experience some atmospheric

drag which slows them down. Fuel is carried to occasionally boost velocity. When fuel runs out - orbital decay + crashing to earth occurs.

18. A boy standing on train station platform observes a NLST (near light-speed transport) train pass through the station. He observes the clocks on the train to be running slower than normal. However, a girl on the train observes the boy's watch, and notices that his watch is running slower than the clocks on the train.

Account for the above situation with reference to the principle of relativity. 3

Both observers see clocks in the other frame run slow. This is because moving clocks run slow, relative to stationary observers. Both the boy & girl's observations are consistent with Special Relativity as Einstein stated that all INERTIAL FRAMES ARE EQUAL (both of these are inertial because neither are accelerating). Therefore both observers would see clocks in the other frame running slow.

17

Marking Criteria	Marks
Gives correct description of change to orbital speed with reason	2
Gives only a general description of the change in orbital speed	1

The satellite's orbital speed decreases with increasing altitude due to reduced centripetal force.

18

Marking Criteria	Marks
Clear, concise discussion, referring to the PoR and frames of reference both being correct as no inertial frame of reference is preferred over another inertial FoR	4
Discussion shows understanding of PoR and mentions frames of reference	3
Reference to PoR and frames of reference	2
Some reference to the PoR	1

19. Michelson and Morley set up an experiment to measure the velocity of Earth relative to the aether. **Mark**

(a) Outline TWO features of the aether model for the transmission of light.

2

1 mark each - any 2 of... filled space, transparent,
permeated all matter, low density, high elasticity
low viscosity, stationary - the absolute rest frame

(b) Recount the Michelson and Morley experiment, which attempted to measure the relative velocity of Earth through the aether, and describe the results they anticipated.

4

Describes the apparatus used - an interferometer with
labels. The use of a coherent light source which would
be split by a half silvered mirror then recombined to
form an interference pattern. M + M expected the interf
pattern to change as the interferometer was rotated
due to the change in relative light speed. Despite good
procedures and repetition no change was ever detected

4

Describes experiment but poorly recounts - expected results
or achieved results or apparatus used

3-2

Has some understanding of the interferometer, or
that light speed was expected to change or
that no result was achieved.

1

20. In your course you performed an investigation to demonstrate the production of an alternating current.

Mark

(a) Describe an experiment you did to produce alternating current, with particular reference to how you verified that alternating current was actually produced.

3

* A description and diagram showing how AC was produced and detected (e.g. oscillating needle)] 3-2

* A description but omits some apparatus or procedural matter] 2-1

* evidence of some knowledge of a suitable exp.] 1

(b) Describe two advantages of using AC generators for large-scale electrical power production. 2

Any 2 reasonable advantages (but must have)
 e.g.: allows voltage to be easily changed
 • no commutator and $\therefore$ better reliability
 • turbines can be spun by a variety of methods
 (wind, water, steam, tidal or wave movement)
 (1 mark each - ONLY FIRST TWO RESPONSES MARKED)

21. The photograph shows a small electrical motor from an electric drill.

3

(a) Name the labelled parts
 A, B, and C and

(b) Describe the function
 Of each

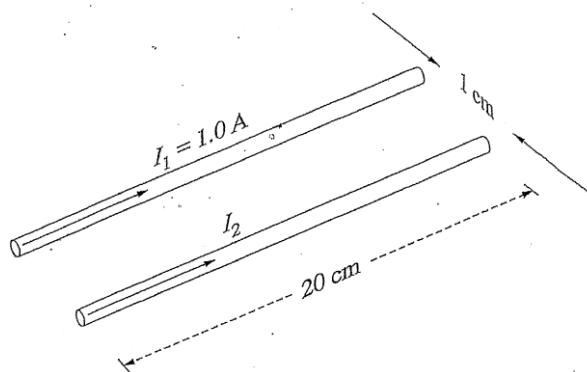
A - Electromagnetic windings - produce a magnetic field

B - Armature windings - carries current

C - Commutator - reverses current each $\frac{1}{2}$ turn

Student Number.....

22. The diagram shows part of an experiment designed to measure the force between two parallel current-carrying conductors. **Mark**
7



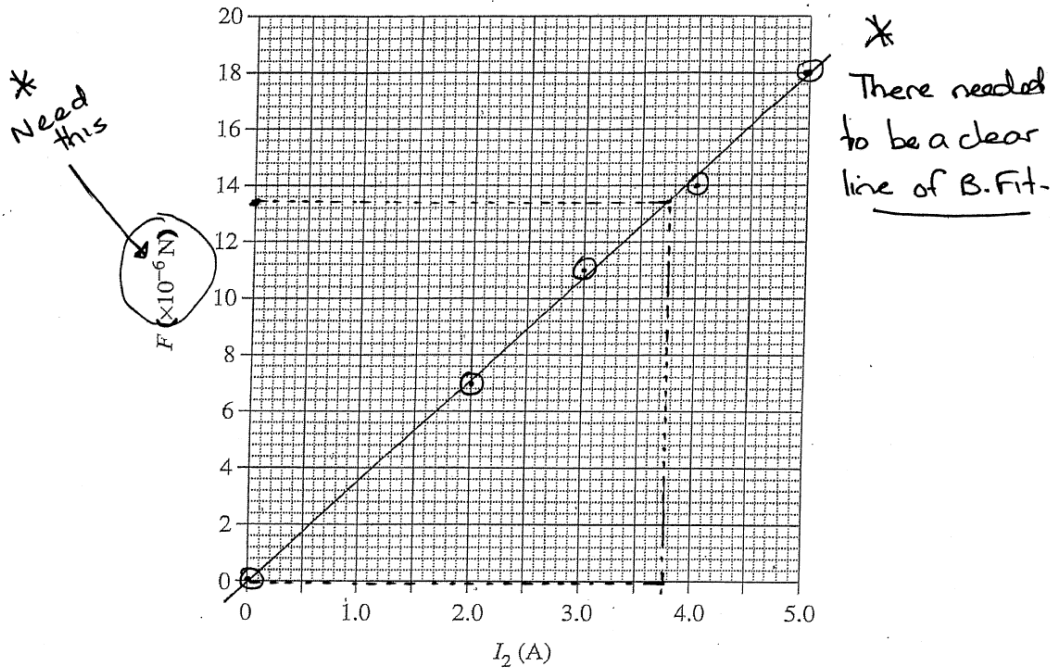
The experimental results are tabulated below.

I_2 (A)	Force ($\times 10^{-6} \text{ N}$)
0	0
2.0	7
3.0	11
4.0	14
5.0	18

Question 22 (continued)

Mark

(a) Plot the data and draw the line of best fit.



(b) Calculate the gradient of the line of best fit the graph.

$$\text{rise} = \frac{(13.4 - 0) \times 10^{-6}}{3.75 - 0} = 3.57 \times 10^{-6} \quad \left(\text{GRADIENT CALCULATIONS MUST BE FROM THE LINE, NOT POINTS} \right)$$

(c) Write an expression for the magnetic force constant k in the terms of the gradient and other 2 variables.

$$F = k I_1 I_2 l \quad \left| \quad F = \frac{k I_1 l}{d} \cdot I_2 \quad \right| \quad \text{gradient} = \frac{k I_1 l}{d} \quad \left| \quad k = \frac{\text{grad} \cdot d}{I_1 \cdot l} \right.$$

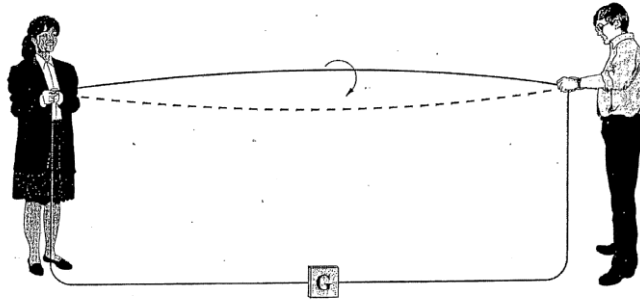
(GRADIENT MUST APPEAR IN THE EXPRESSION) $\therefore k = \text{grad} \times 0.05$

(d) Use this expression and the gradient calculated in part (b) to determine the value of the magnetic force constant k .

$$k = \text{gradient} \times 0.05 = 1.79 \times 10^{-7}$$

Student Number.....

23. In a particular experiment a long length of copper wire of very resistance is rotated by two students. The ends of the wire are connected to a galvanometer G, and a current is detected. Marks 4



Explain the effect of increasing the speed of rotation on the current measured by the galvanometer

Outline/Identify that more speed gives a greater rate of cutting field lines or greater rate of change of magnetic flux. States Faraday law or that induced emf = $\frac{\Delta \text{FLUX}}{\Delta \text{time}} = \frac{B \Delta A}{t} = Blv$. Explains increased emf occurs and $\therefore$ more current. } 4 marks

Relates speed to flux change and $\therefore$ greater current } 3-2

Identifies more current will be read on galvanometer] 1

- *BANDS AND HOLES 24. Using silicon as an example of a semiconductor, describe how it carries a current and how doping affects the process. 2

MUST BE REFERRED TO - Describes the movement of electrons and holes in silicon and that doping increases the number of available electrons or holes OR that doping changes/reduces the energy required to make the semiconductor conductive OR that doping changes electrical properties } 2-3

- Describes charge movement OR doping OR a dodgy description of both } 1

thought experiment
not acceptable

Student Number.....

25. During your course you carried out an investigation to model behaviour of semiconductors. Marks including the concept of holes.

* NEED³

Outline what you did in your investigation. Explain how the model showed conduction in semiconductors.

Gives apparatus used (or diagram) and how it shows holes/electrons being able to move with energy input to cross F.E. gap i.e. Valence band → Conduction band } 3-2

Describes apparatus with little procedural info OR doesn't refer to hole or electron movement (i.e. how it occurs) } 1

E.g. petri dish, marbles. Marble represent electrons, when shaken (E in) CONDUCTION BAND VALENCE BAND } marbles jump up (conduction band) and leave a HOLE. Both the marble + hole can move } 3

26. In early studies, the observed characteristics of cathode rays led to the belief that they were electromagnetic waves.

Describe the wave-like properties of cathode rays and explain how other evidence shows them to be particles.

EXPLANATION OF PROPERTIES REQUIRED - A LIST IS NOT ENOUGH } Gives 2 wave-like props. (travels in straight lines, fluoresces green light, not deflected by E fields in 1880) and at least 1 particles property e.g. S - imparts momentum, stopped by thin metals, deflected by B fields PLUS EXPLANATION } 3

Gives 1 wavelike + 1 particle property } 2

Gives only wave OR particle props. } 1

27. A physics student was conducting an investigation on the photoelectric effect. The student used an infrared laser with a wavelength of 1.55×10^{-6} m for this investigation.

(a) Calculate the energy of a photon from this laser.

$E = h\nu = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{1.55 \times 10^{-6}} = 1.28 \times 10^{-19} \text{ J}$ } 2

NO UNIT = -1

(b) When the laser light was shone onto a photo-cell, no current was detected. The student

3

increased the intensity of the light but still detected no current. OR WORK FUNCTION
Explain this observation.

clearly relates ^{answer} to threshold frequency explaining that } 3 mks
increasing intensity doesn't increase photon ϵ so
no photo electrons will be ejected.

Understands concept of threshold frequency to some } 2 - 1
degree

28. (a) Calculate the frequency of a photon of blue light of wavelength 460nm.

1

$$f = \frac{c}{\lambda} = \frac{3 \times 10^8}{460 \times 10^{-9}} = 6.52 \times 10^{14} \text{ Hz}$$

(CALCULATION MUST BE CORRECT AND COMPLETE)

(b) Identify Planck's hypothesis that allowed him to successfully account for the black body radiation curve.

1

* BOTH REQUIRED
Energy is absorbed and emitted by atomic oscillators in black
body's in packets of energy called quanta.

29. Outline how Hertz measured the speed of radio waves.

3

* Describes the apparatus with acceptable accuracy (1 mark)

* Outlines the connection of the transmitter + receiver
with a wire (1 mark)

* demonstrate an understanding that an
interference effect occurred from which (1 mark)
Hertz used $c = f\lambda$ to determine "c"

30. Describe how superconductors and magnetic levitation have played a part in the development of the maglev train.

2

NB - NOT MEISSNER EFFECT

MUST
SHOW
DEPTH OF
KNOWLEDGE

DESCRIBES THIS
- Superconductors have led to much stronger electromagnets
- Superconducting electromagnets support and propel maglev
trains, (diagram optional but useful.)

missing one area (of the 2) noted above

From Quanta to Quarks. (25 Marks)

Marks

Must
describe, then
give FOR
AGAINST
argument

(a) Discuss Rutherford's proposal of the nuclear atom with orbiting electrons

4

① Describes Nuclear atom (diagram will do if labelled).....

② Give experimental evidence for at least 1 feature

③ Gives at least 1 argument against the model e.g.

no emc detected from accelerating electrons

4 - 3

Describes nuclear atom but missing evidence for OR
against

3 - 2

Describes Nuclear atom

1

(b) In refining the model of the atom, Bohr began with three postulates.
State 2 of Bohr's postulates.OR "stationary
States" 2

ANY 2 OF - (ONLY FIRST 2 are marked)

1) orbiting electrons exist on quantised energy levels. Electrons
on these levels are stable and emit no energy2) Energy is absorbed or emitted when electrons change
E level according to $hf = E_1 - E_2$

3) Angular momentum of electrons is quantised

(c) Identify experimental evidence that supported one of Bohr's postulates and explain
how it provided this support.

2

Correct identification of relevant evidence (1 mark)

Justifies how the stated evidence supports the
particular evidence (1 mark)

* De Broglie only provides indirect support for 1ST POSTULATE - 1 mark

* Davidson and Germer accepted ONLY IF linked to 3RD POSTULATE
and explained with algebraic prove.

(d) Define the term transmutation.

1

The transformation of 1 element into another by the
bombardment of nuclei with particles.

Student Number.....

Marks

(e) As a result of the studying the electrons emitted during beta decay, Pauli suggested the existence of a then unknown particle. Discuss Pauli's suggestion, and relate this to the energy of the emitted electrons. 4

- 4 marks {
- Outlines what the suggestion of Pauli's was ("NEUTRINO" was not used by Pauli)
 - Gives reasons for the suggestion relating to β particle KE's + cons. of p.
 - Gives reasons against the proposal - i.e. no experimental detection

- 3 marks {
- As above but omits or communicates one of the points poorly

- 2 marks {
- Demonstrates some understanding of Pauli's suggestion

- 1 {
- Some attempt

(f) Calculate the mass of a particle that has a De Broglie wavelength of 2.5×10^{-12} m when moving with speed of 200 ms^{-1} . 1

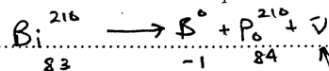
$$\lambda = \frac{h}{mv}, \quad m = \frac{h}{\lambda v}$$

$$= \frac{6.63 \times 10^{-34}}{2.5 \times 10^{-12} \times 200}$$

$$= 1.33 \times 10^{-24} \text{ kg}$$

- Correct working essential {
- (g) Calculate the energy of the lowest frequency photon emitted in the Balmer series? 2
- lowest f when $n=2, n_1=1$
- $$\frac{1}{\lambda} = 1.097 \left(\frac{1}{2^2} - \frac{1}{1^2} \right)$$
- $$\lambda = 9.116 \times 10^{-8} \text{ m}$$
- $$f = \frac{c}{\lambda} = \frac{3 \times 10^8}{9.116 \times 10^{-8}} = 3.291 \times 10^{15} \text{ Hz}$$
- $$E = hf = 3.06 \times 10^{-19} \text{ J}$$

(h) Write the transmutation equation for the beta decay of Bi_{83}^{210} . 2



(or ν)

* -1 mark for EACH missing or incorrect part

For full marks - MUST MAKE

Student Number.....

EXPLICIT statement about the "SIGNIFICANCE OF THE CONSERVATION LAWS"

Marks

(i) Explain the significance of the conservation laws, in Chadwick's discovery of the neutron. 4

*

4-3

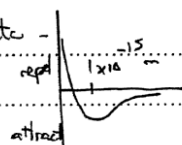
1. Describes the Rutherford + Geiger expt resulting in protons being ejected from paraffin.
2. Explains that the Law of Cons. of momentum could not be satisfied unless the highly penetrative projectile was much more massive than initially suggested
3. Explains that conservation of mass and charge suggested particle had to be similar in mass to a proton and have no charge
4. EXPLICIT STATEMENT OF SIGNIFICANT OF CONSERVATION LAWS

2-1

Identifies conservation of momentum as being relevant

(j) Justify the existence of the strong nuclear force. 3

- Describes nature of nuclear force, range etc -
- that is very short range



- Outlines gravitational and electrostatic forces and relative sizes of each (Electrostatic repulsion is much greater)
- explains need for S.N.F. to overcome electrostatic repulsion in stable nuclei.

END OF EXAMINATION

Describes the nature of SNF and refers to strong electrostatic forces

Some idea